DATA SCIENCE CASE STUDY

Bike Sharing Demand: Should We Buy More Bikes?

Data quality, hypothesis testing, demand model and a natural-language prototype for operations

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INTRODUCTION

The business problem

THE CEO ASKS

“Riders can’t find a bike, yet bikes sit unused. Should we just buy more bikes?”

Dataset: 17,379 hourly rentals, Jan 2011 to Dec 2012, with weather and calendar fields.

1

Quality checks and features

2

One testable hypothesis

3

Model to estimate demand

4

Natural-language prototype

CONTENT

Solution flow

01

Data quality

Missing values, duplicates, timestamp gaps

02

Feature engineering

Rush hour, weekend, cyclical time, real units

03

Exploration

Hour, season, weather and working-day patterns

04

Hypothesis test

Are working-day peaks sharper? Mann–Whitney U

05

Demand model

Random Forest with chronological split

06

Capacity and prototype

Percentile benchmark and plain-English questions

STEP 1

Data quality checks

17,379
hourly records

0
missing cells or duplicates

0
rows where casual + registered ≠ cnt

165
missing hourly timestamps

```
hour["timestamp"] = hour["dteday"] + \
    pd.to_timedelta(hour["hr"], unit="h")
expected = pd.date_range(start, end, freq="h")
missing = expected.difference(hour["timestamp"])
# 17,544 expected vs 17,379 available
```

Decision

No blank cells, yet 165 whole hours are absent. They were not filled with zero, because a missing record does not mean zero rentals.

Data was only de-duplicated and sorted by time.

STEP 2

Feature engineering

```
hour["is_rush_hour"] = (
    hour["hr"].isin([7,8,9,16,17,18,19])
    & (hour["workingday"] == 1)).astype(int)
hour["hour_sin"] = np.sin(2*np.pi*hour["hr"]/24)
hour["hour_cos"] = np.cos(2*np.pi*hour["hr"]/24)
hour["temp_c"] = hour["temp"] * 41
```

is_weekend, is_rush_hour
Commute peaks on working days

hour / month sin and cos
Midnight sits next to 11 PM

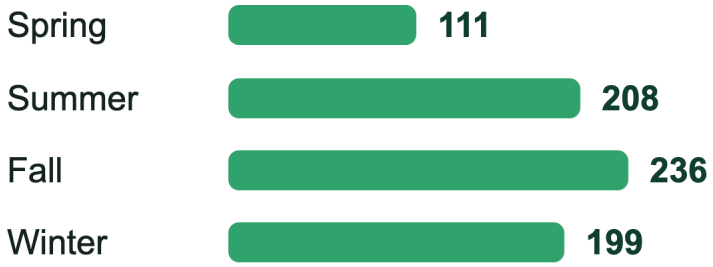
temp_c, humidity_pct, wind_kmh
Real-world units to interpret

casual and registered excluded
They sum to cnt, so they would leak the target

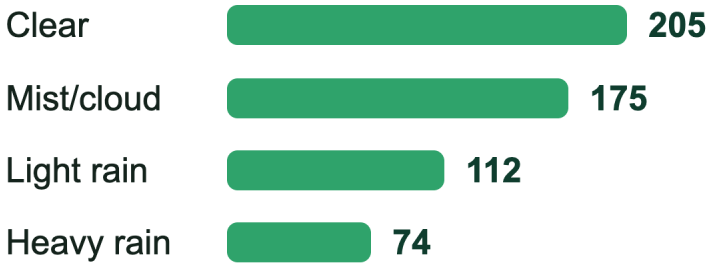
STEP 3

What drives demand

Average rentals by season



Average rentals by weather



2012 working-day peaks, avg rentals per hour



Takeaway

Demand swings with time of day, season and weather, so a single fleet number hides when bikes are actually short.

STEP 4

Hypothesis: are working-day peaks sharper?

H0

Working days do not have a higher peak-to-average demand ratio.

H1

Working days have a higher peak-to-average ratio (max hourly ÷ mean hourly, per complete day).

2.96

working days (443)

2.30

non-working days (212)

6.6e-79

one-sided p-value

```
u, p =
stats.mannwhitneyu(
    working,
non_working,
    alternative="greater
")
# p < 0.05 → reject H0
```

Result: peaks are sharper on working days. This is a demand pattern, not proof that redistribution beats buying; that needs a field experiment.

STEP 5

Demand model: Random Forest

Setup

- 200 trees, min_samples_leaf = 2
- Train before 7 Aug 2012 (13,891 rows)
- Test on later data (3,488 rows)
- Time-ordered split, no random shuffle

Top features

hour 37% · rush hour 13% · feel-like temp 10% · year 8% · temp 7%

45.09

MAE, rentals per hour

69.57

RMSE

0.90

R^2 on unseen data

95.39

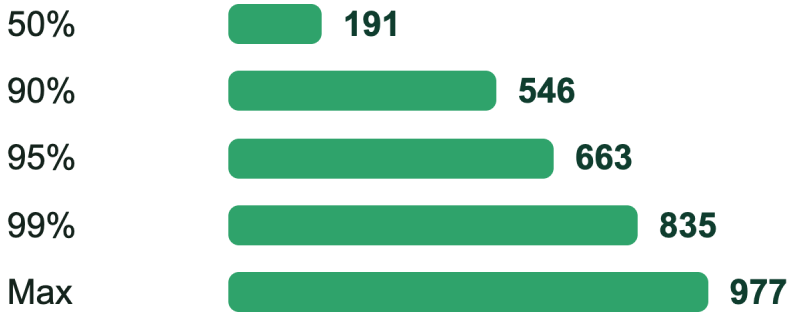
baseline MAE (season, day type, hour)

The model cuts error by about 53% versus the historical-average baseline.

STEP 6

Capacity benchmark and prototype

2012 hourly demand percentiles



A planning benchmark in rentals per hour, not an exact physical fleet size.

Ask in plain English

“How many rentals at 8am on 2012-07-15?”
→ 136 rentals

“…and if it rains?”
→ 133 rentals

“Peak hours on 2012-07-15?”
→ 1 PM 488, 12 PM 486, 2 PM 484

“Bikes needed at 35 degrees?”
→ Busiest hour 2 PM, 527

ask_manager() reads date, weather, temperature and hour, then the model predicts all 24 hours.

CONCLUSION

Redistribute first, then buy selectively

What the data shows

- Demand depends strongly on hour, season and weather
- Working-day peaks are significantly sharper ($p \approx 7e\text{-}79$)
- Model explains 90% of variation, MAE 45 per hour
- 95th percentile benchmark: 663 rentals per hour

Recommended pilot

- A: current operations
- B: redistribute bikes before peaks
- C: add bikes at high-demand spots
- Run 4+ weeks; compare failed rentals, complaints, cost, dock-full events

Data still needed for an exact fleet size: station inventory, trip duration and failed-rental requests.